

SINGLY LINKED LIST

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

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QUESTIONS 10 / 10 completed

Linear Search

LINEAR SEARCH

Create a C program to perform Linear Search on an array of 10 integers. The program should allow manual input or random generation of array elements. It must display the first position of the element (1-based indexing), total

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <time.h>
4
5 int main() {
6     int arr[10];
7     int choice, i, key, comparisons = 0;
8     int position = -1;
9
10    printf("Choose input method:\n");
11    printf("1. Manual Input\n");
12    printf("2. Random Generation\n");
13    printf("Enter your choice: ");
14    scanf("%d", &choice);
15
16    if (choice == 1) {
17        printf("Enter 10 integers:\n");
18        for (i = 0; i < 10; i++) {
19            printf("Element %d: ", i + 1);
20            scanf("%d", &arr[i]);
21        }
```

INPUT

1
20 30 40 50
30

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QUESTIONS 10 / 10 completed

Binary Search

BINARY SEARCH

"Write a C program implementing binary search on a sorted array of 15 integers. The program must: (1) accept an unsorted array and sort it first, (2) perform binary search, (3) display search iterations with mid-point calculations, (4)

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 void bubbleSort(int arr[], int n) {
4     for (int i = 0; i < n - 1; i++) {
5         for (int j = 0; j < n - 1 - i; j++) {
6             if (arr[j] > arr[j + 1]) {
7                 int temp = arr[j];
8                 arr[j] = arr[j + 1];
9                 arr[j + 1] = temp;
10            }
11        }
12    }
13 }
14
15 int isSorted(int arr[], int n) {
16     for (int i = 0; i < n - 1; i++) {
17         if (arr[i] > arr[i + 1]) return 0;
18     }
19     return 1;
20 }
```

INPUT

20 30 40 50 60
30



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QUESTIONS 10 / 10 completed

Binary Search- Iterc

BINARY SEARCH-ITERATIVE

Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day >=30°C and last day <=25°C. Display temperatures, indices (0-based), or ""Not found"". Validate sorted; n<=50.

PROGRAM EDITOR

C

Run

Save

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     int n;
6     cin >> n;
7
8     vector<int> a(n);
9     for (int i = 0; i < n; i++) cin >> a[i];
10
11     sort(a.begin(), a.end());
12
13     // First number >= 30 (lower_bound style) - iterative binary search
14     int target1 = 30;
15     int lo = 0, hi = n - 1;
16     int ans1 = -1; // index of first >= 30
17
18     while (lo <= hi) {
19         int mid = lo + (hi - lo) / 2;
20         if (a[mid] >= target1) {
21             ans1 = mid;
```

INPUT

Your INPUT goes here!

36°C Partly sunny

Q Search

ENG IN



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QUESTIONS 10 / 10 completed

Linear Search -Sort

LINEAR SEARCH - SORT

Read n book page counts into an array. Sort ascending using any method. Use linear search to find the first book having >= 300 pages and the last book having <= 150 pages. Display page counts and indices (0-based), or ""Not found"". Validate

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5
6     if (scanf("%d", &n) != 1) return 0;
7     if (n <= 0 || n > 50) {
8         return 0;
9     }
10
11     int pages[50];
12
13     for (int i = 0; i < n; i++) {
14         scanf("%d", &pages[i]);
15     }
16
17     for (int i = 0; i < n - 1; i++) {
18         for (int j = 0; j < n - i - 1; j++) {
19             if (pages[j] > pages[j + 1]) {
20                 int temp = pages[j];
21                 pages[j] = pages[j + 1];
```

INPUT

Your INPUT goes here!



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QUESTIONS 10 / 10 completed

Linear Search - Ana

LINEAR SEARCH - ANAGRAMS

"Read n strings, use linear search to find all anagrams of a target string (same chars, any order). Display positions (1-based) and matching strings. Case-insensitive; n<=50. Sample Input: 5 1) listen silent 2) race care

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int isAnagram(char *s1, char *s2) {
6     int len1 = strlen(s1);
7     int len2 = strlen(s2);
8     if (len1 != len2) return 0;
9
10     int count[256] = {0};
11
12     for (int i = 0; i < len1; i++) {
13         count[(unsigned char)tolower(s1[i])]++;
14         count[(unsigned char)tolower(s2[i])]--;
15     }
16
17     for (int i = 0; i < 256; i++) {
18         if (count[i] != 0) return 0;
19     }
20     return 1;
21 }
```

INPUT

Your INPUT goes here!

QUESTIONS 10 / 10 completed

Linear Search on Array

LINEAR SEARCH ON ARRAY

"Develop a C program to perform Linear Search on an array of N integers. Accept array size dynamically"

Sample Input/Output
Enter size: 5
Array: 10 20 30 40 50
Enter target: 40
Comparison Trace:
10 ≠ 40

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main() {
5     int n, target, i, found = 0, comparisons = 0;
6
7     printf("Enter size: ");
8     if (scanf("%d", &n) != 1) return 1;
9
10
11     int *arr = (int *)malloc(n * sizeof(int));
12     if (arr == NULL) {
13         printf("Memory allocation failed!\n");
14         return 1;
15     }
16
17     printf("Array: ");
18     for (i = 0; i < n; i++) {
19         scanf("%d", &arr[i]);
20     }
21
```

INPUT

Your INPUT goes here!

QUESTIONS 10 / 10 completed

Binary Search - Arr

BINARY SEARCH - ARRAY

Write a C program to perform Binary Search on a sorted array of 10 integers. The program should accept the array elements and a target value from the user. It must display the position of the element and the total number of comparisons made.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[10];
5     int target, low, high, mid, i;
6     int comparisons = 0;
7     int found = 0;
8
9     printf("Enter 10 sorted integers:\n");
10    for (i = 0; i < 10; i++) {
11        scanf("%d", &arr[i]);
12    }
13
14    printf("Enter element to search: ");
15    scanf("%d", &target);
16
17    low = 0;
18    high = 9;
19
20    while (low <= high) {
21        comparisons++;

```

INPUT

Your INPUT goes here!

QUESTIONS 10 / 10 completed

Linear Search -Tem

LINEAR SEARCH - TEMPERATURE

"Read n daily temperatures into an array. Use Linear Search to find the last day whose temperature is less than 20°C. Display temperatures, index (0-based), and value. If not found, print ""Not found"". Validate n ≤ 50."

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     int temps[50];
6     int lastIndex = -1;
7
8     if (scanf("%d", &n) != 1) {
9         return 0;
10    }
11
12    if (n > 50 || n <= 0) {
13        return 0;
14    }
15
16    for (i = 0; i < n; i++) {
17        scanf("%d", &temps[i]);
18    }
19
20    for (i = 0; i < n; i++) {
21        printf("%d", temps[i]);

```

INPUT

Your INPUT goes here!



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QUESTIONS 10 / 10 completed

Linear Search -Price

LINEAR SEARCH - PRICES

"Read n product prices into an array. Sort ascending using any method. Use linear search to find the first price >= ₹500 and the last price <= ₹200. Display prices and indices (0-based), or ""Not found"". Validate sorted order; n<=50."

PROGRAM EDITOR

C

Run
Save

```

1 #include <stdio.h>
2
3 int main() {
4     int n;
5     if (scanf("%d", &n) != 1) return 0;
6
7     if (n < 1 || n > 50) {
8         printf("Invalid n\n");
9         return 0;
10    }
11
12    int a[50];
13    for (int i = 0; i < n; i++) {
14        scanf("%d", &a[i]);
15    }
16
17    for (int i = 0; i < n - 1; i++) {
18        for (int j = 0; j < n - 1 - i; j++) {
19            if (a[j] > a[j + 1]) {
20                int temp = a[j];
21                a[j] = a[j + 1];
22                a[j + 1] = temp;
23            }
24        }
25    }
26
27    // Linear Search for first price >= 500 and last price <= 200
28    int first = -1, last = -1;
29    for (int i = 0; i < n; i++) {
30        if (a[i] >= 500) first = i;
31        if (a[i] <= 200) last = i;
32    }
33
34    if (first == -1 || last == -1) {
35        printf("Not found\n");
36    } else {
37        printf("First price >= 500 at index %d\n", first);
38        printf("Last price <= 200 at index %d\n", last);
39    }
40
41    return 0;
42 }
```

INPUT

Your INPUT goes here!



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QUESTIONS 10 / 10 completed

Bin. Search-Iterative

BIN. SEARCH-ITERATIVE-TEMP

"Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day >= 30°C and last day <= 25°C. Display temperatures, indices (0-based), or ""Not found"". Validate sorted; n<=50."

PROGRAM EDITOR

C

Run
Save

```

1 #include <stdio.h>
2
3 void sortArray(int arr[], int n) {
4     int i, j, temp;
5     for (i = 0; i < n - 1; i++) {
6         for (j = 0; j < n - i - 1; j++) {
7             if (arr[j] > arr[j + 1]) {
8                 temp = arr[j];
9                 arr[j] = arr[j + 1];
10                arr[j + 1] = temp;
11            }
12        }
13    }
14 }
15
16 int findFirstGreaterEqual(int arr[], int n, int target) {
17     int low = 0, high = n - 1, result = -1;
18     while (low <= high) {
19         int mid = low + (high - low) / 2;
20         if (arr[mid] >= target) {
21             result = mid;
22             high = mid - 1;
23         } else {
24             low = mid + 1;
25         }
26     }
27     return result;
28 }
```

INPUT

Your INPUT goes here!